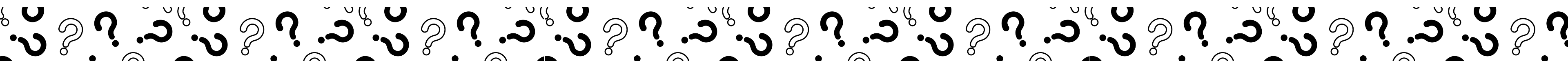


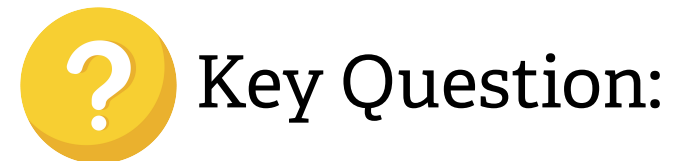


Popular but Underperforming

Understanding Mismatch Between Course Ratings and Student Engagement

Presented by: Abdul Farooqu Karim Kutty
Master of Data Science, Macquarie University





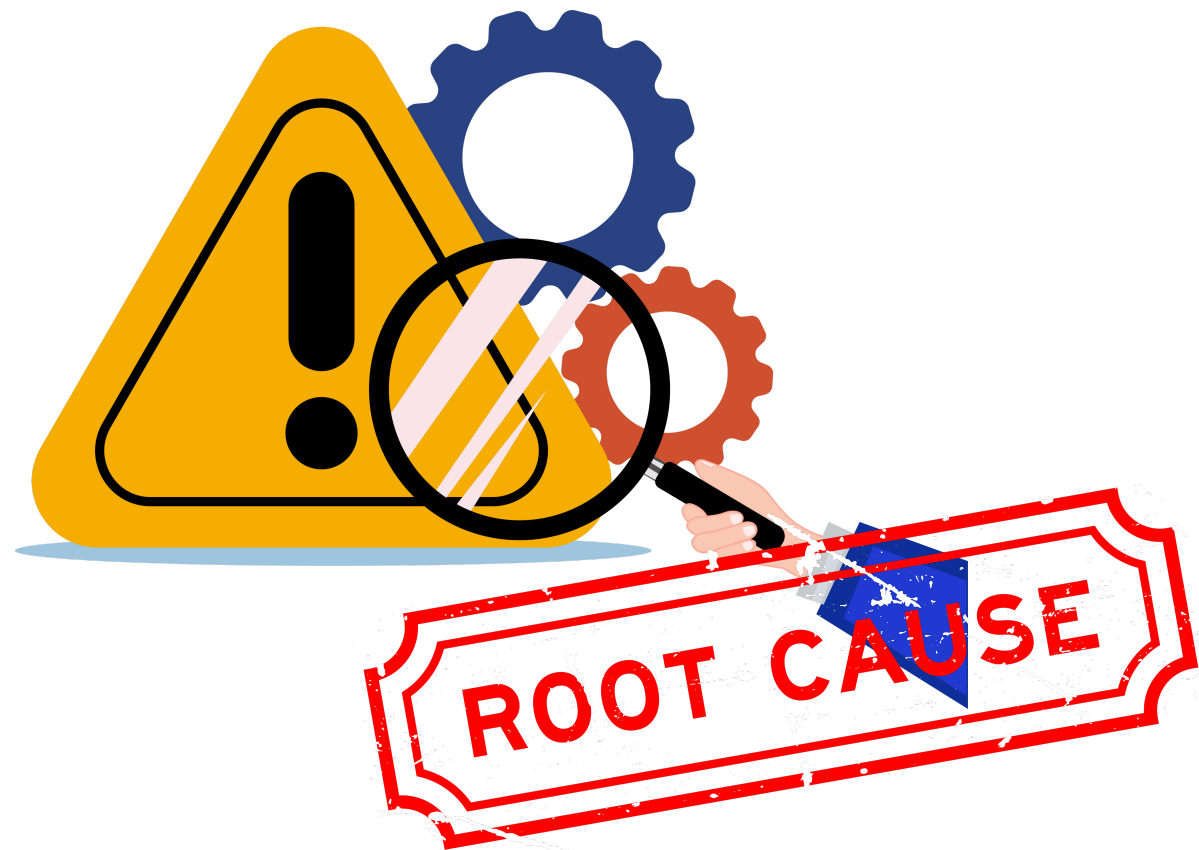
Why do some highly-rated courses have low student engagement or enrollment?

Why This Matters

- Course ratings are often seen as a proxy for success.
- But high ratings don't always lead to real engagement.
- This mismatch can lead to **misguided decisions** about course quality, resource allocation, or marketing.

Real-World Impact

- University Strategy: Misinterpreted data may cause the university to invest in or **promote courses** that don't meet student needs
- Content Development: Course teams might focus on the wrong success metrics.
- Perception vs Reality: What students say they like $\neq$ what they actually pursue.



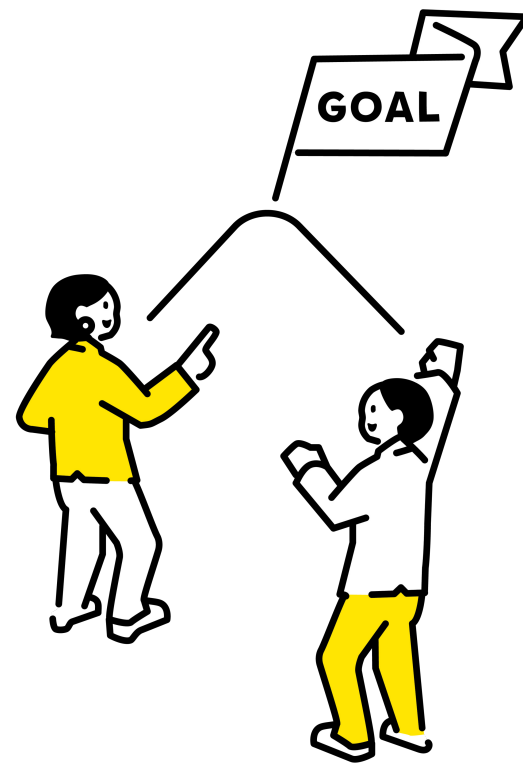
Root Cause Focus

We explore the "**why**" behind this mismatch to move beyond surface-level metrics and improve evidence-based academic planning.

Root Cause Thinking

What is Root Cause Analysis?

“Not just what’s happening , but why it’s happening.”



Our Goal:

Move beyond surface observations (e.g., “This course has low enrollments”) → to understanding the underlying drivers (e.g., difficulty level, certificate value, organization reputation).

Why This Approach?

Helps connect **sentiment with behavior**

Identifies hidden patterns behind student decisions

Supports targeted interventions, not just assumptions

How We Frame the Problem:

We treat this as a perception–behavior gap:

- Ratings reflect student sentiment
- Enrollments reflect actual student action

We ask: Where do they misalign and why?

Dataset & Methodology

- **Dataset & Methodology**

Source: Structured dataset of 800+ Coursera courses

Includes:

Course title

Rating (out of 5)

Enrollment count

Difficulty level

Certificate type

Course organization

- **Data Cleaning & Preparation**

- **Converted enrollment text (e.g., “1.2k”)**

- **Numeric**

- **Removed non-English course titles**

- **One-hot encoded categorical features:**

Difficulty (Beginner, Intermediate, Advanced)

Original DataFrame:

User_ID	Difficulty	Certificate_Type	Organization	Score
1	Beginner	Course Completion	Macquarie University	85
2	Intermediate	Professional Certification	Coursera	92
3	Advanced	Course Completion	edX	78
4	Beginner	Micro-credential	Macquarie University	90

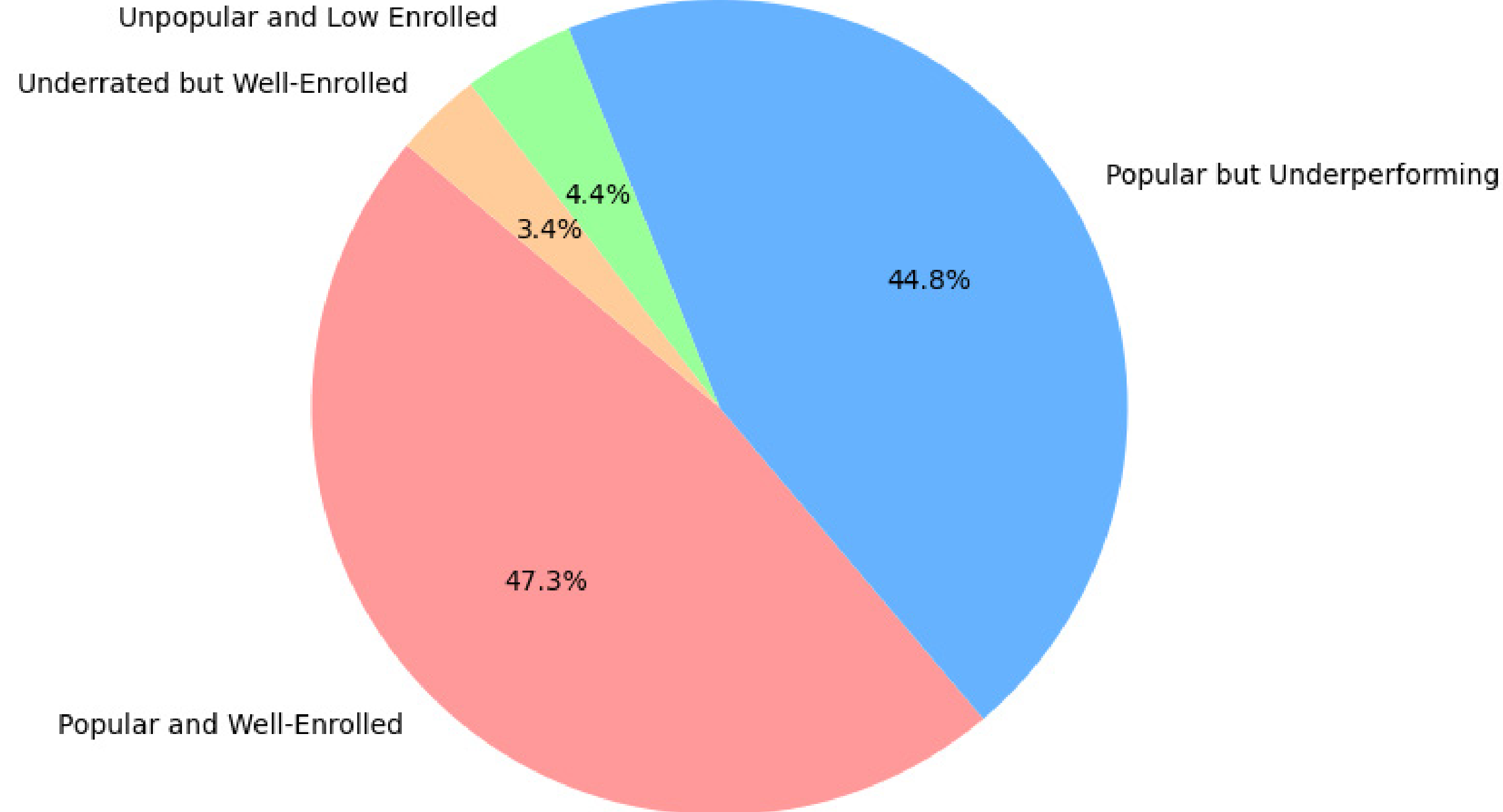


Initial Descriptive Insights

*A high course rating does **not always** lead to **high student enrollment**.*

*This signals a behavioural **mismatch** strong sentiment doesn't guarantee strong engagement.*

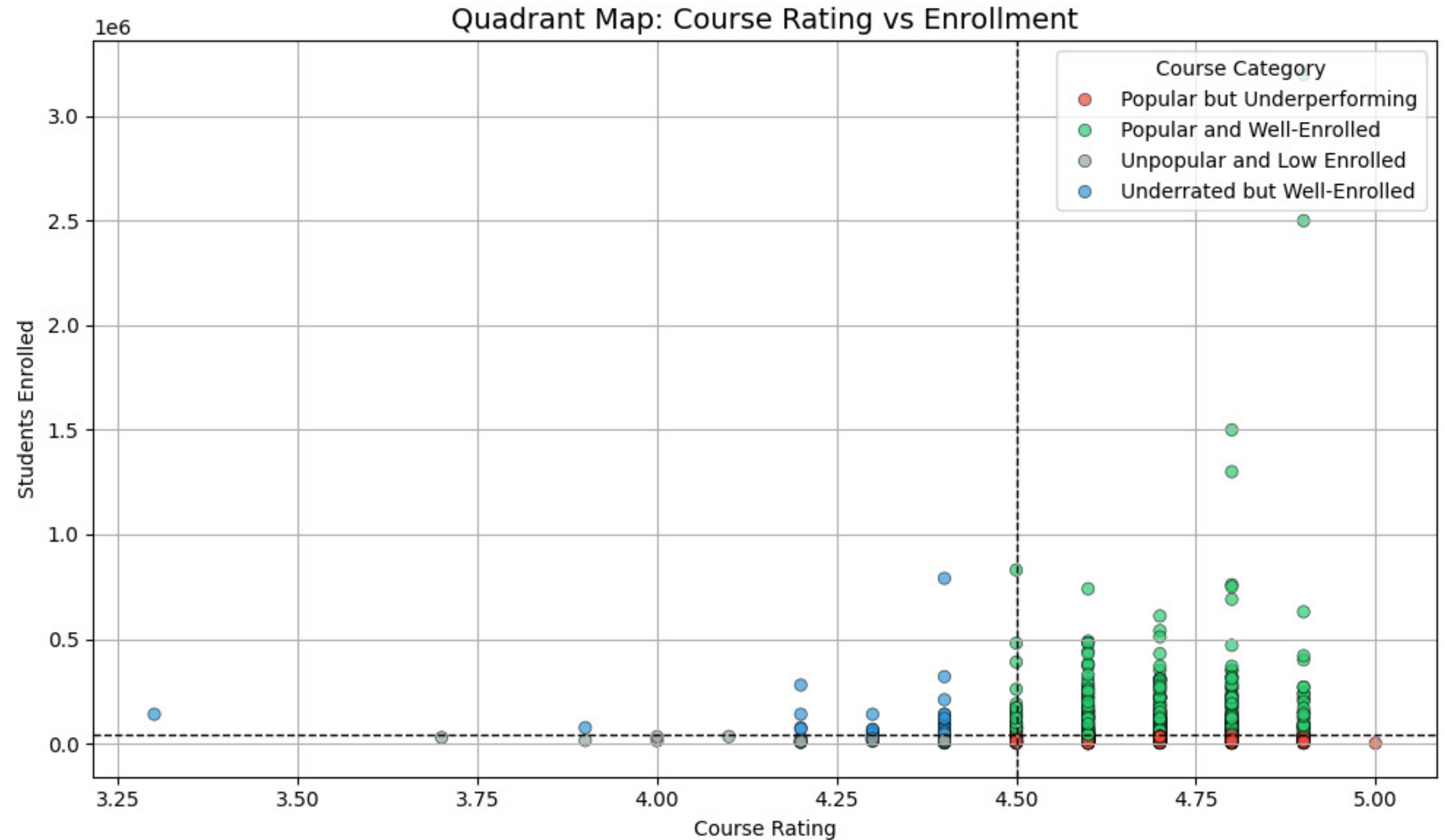
Course Distribution by Rating vs Enrollment Quadrant



Quadrant Map

*Popular but Underperforming
→ These are well-rated but under-enrolled courses,
the main focus of our root cause analysis.*

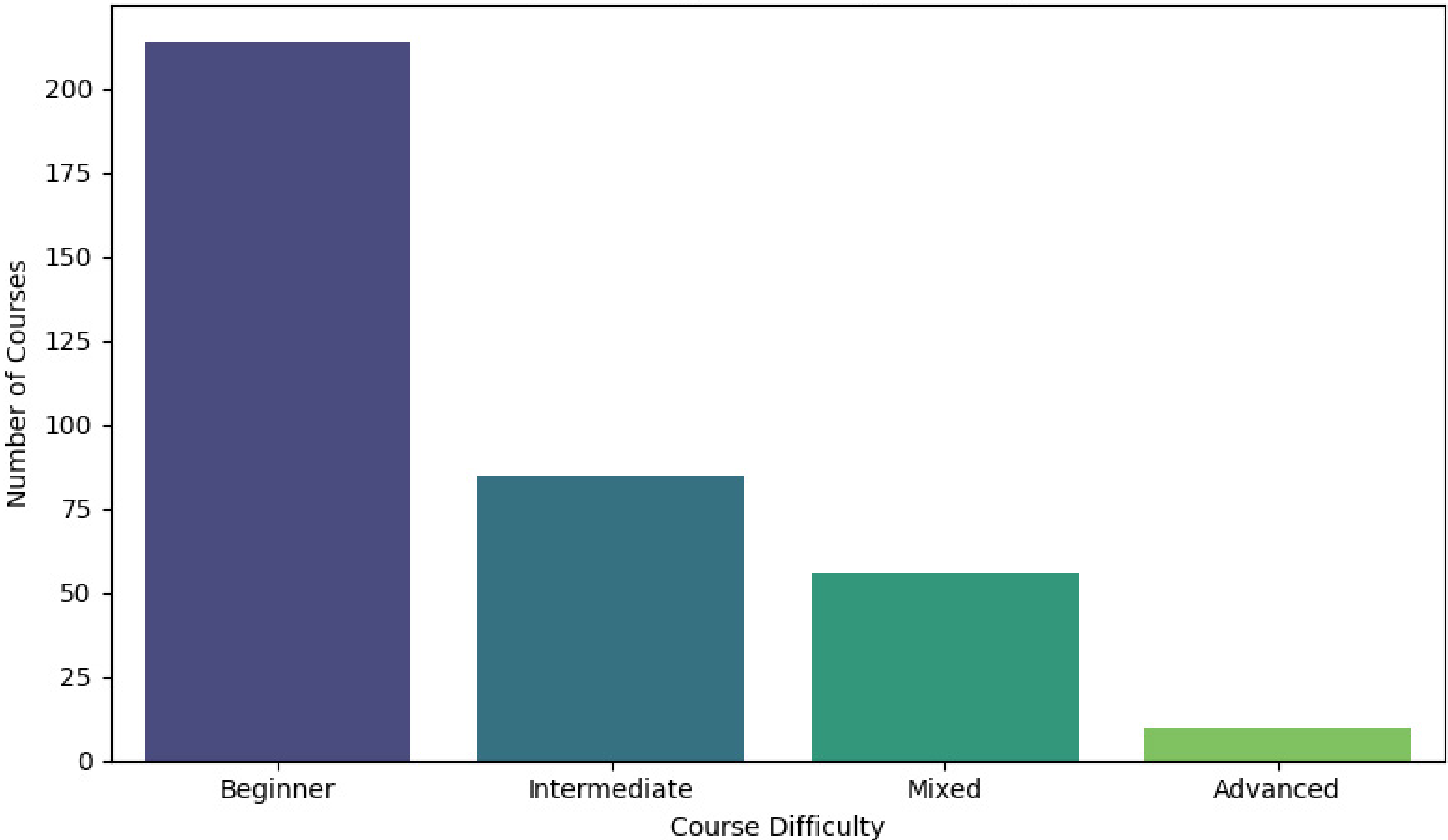
Course Ratings vs Student Enrollments



We discovered that a 5-star course isn't always popular ,the 'why' led us to **dig deeper into other course attributes.**

Root Cause Exploration

Underperforming Courses by Difficulty Level



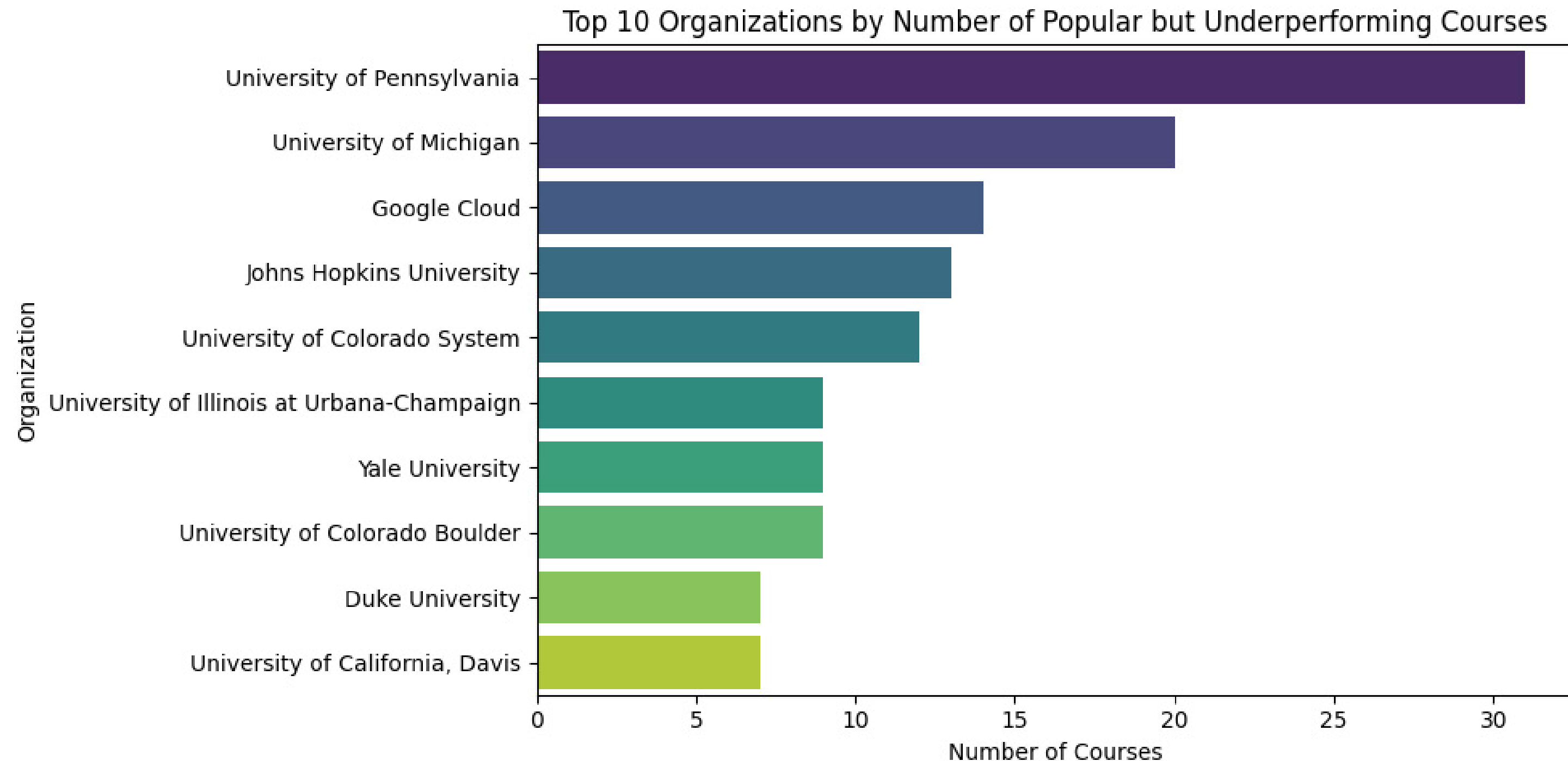
What We Investigated:

Among the Popular but Underperforming courses, we explored:

Course difficulty can act as a barrier, even if users rate it highly after completing it.

Organisation vs popular but underperforming courses

Prestigious universities dominate the list of popular courses with poor learning outcomes, revealing a **gap between brand reputation and educational effectiveness**.



Difficulty vs Certificate Type

– Enrollment Breakdown



Difficulty vs Certificate Type (Mismatch Courses)

course_Certificate_type	COURSE	PROFESSIONAL	CERTIFICATE	SPECIALIZATION	Total
course_difficulty					
Advanced	7		0	3	10
Beginner	133		1	80	214
Intermediate	51		0	34	85
Mixed	56		0	0	56
Total	247		1	117	365

Key Insight:

“Beginner-level courses offered by top organizations were often highly rated, but had fewer enrollments, especially when certificates were not offered or were costly.”

Why This Matters:

Students may avoid courses that feel too basic or don’t offer tangible value (e.g., certificate).
Highlights a gap between sentiment (good rating) and motivation (what drives action).

Can We Predict Course Enrollments from Features?



Goal:

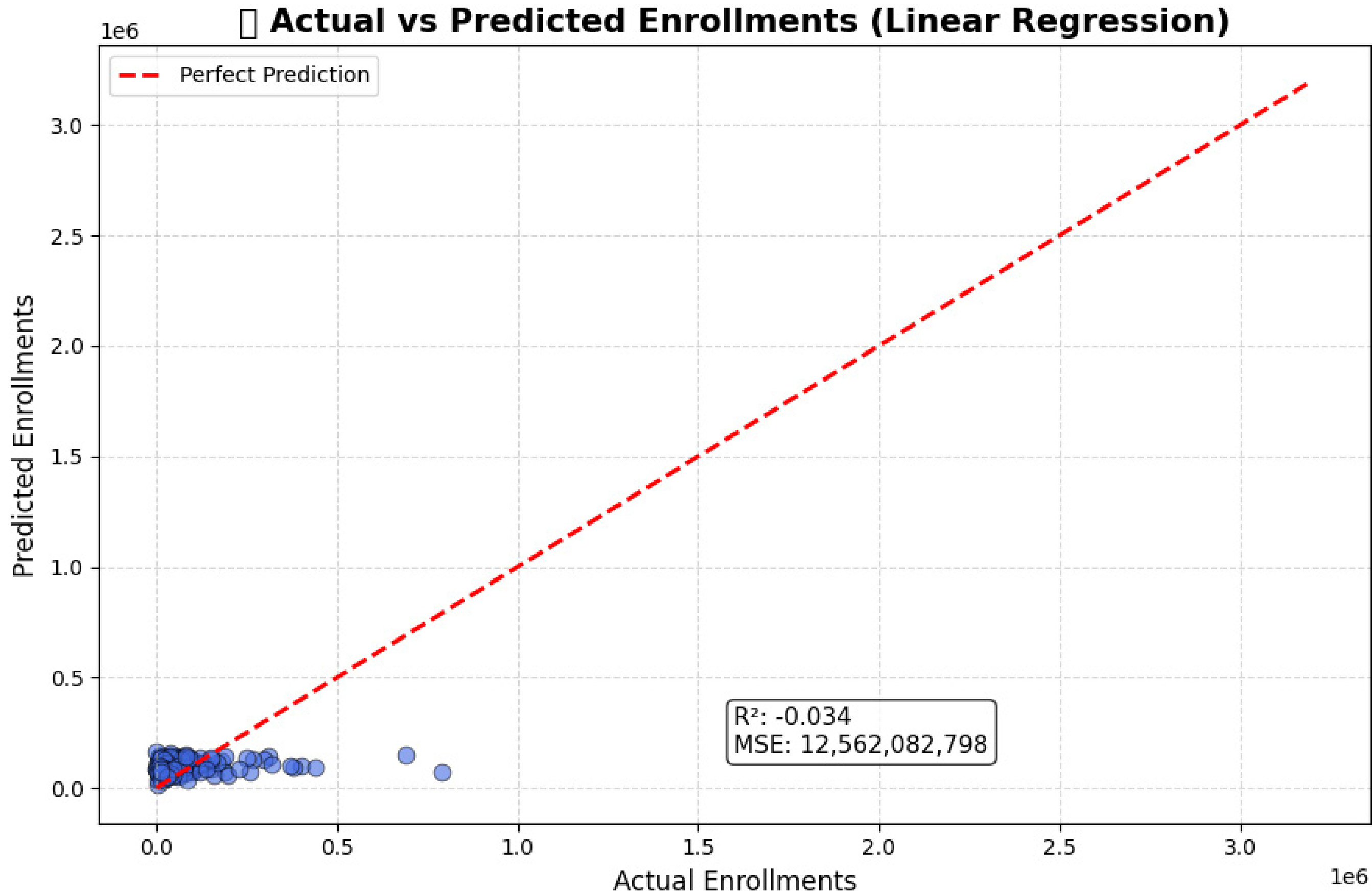
To explore whether course-level features (like rating, difficulty, certificate type) can predict student enrollment.

- **Models Used:**
Random Forest Regressor
Linear Regression

- **Key Insight:**

“Even highly rated courses didn’t guarantee enrollments.”

The models gave low R^2 scores, showing that observable features alone couldn’t explain student behavior.”

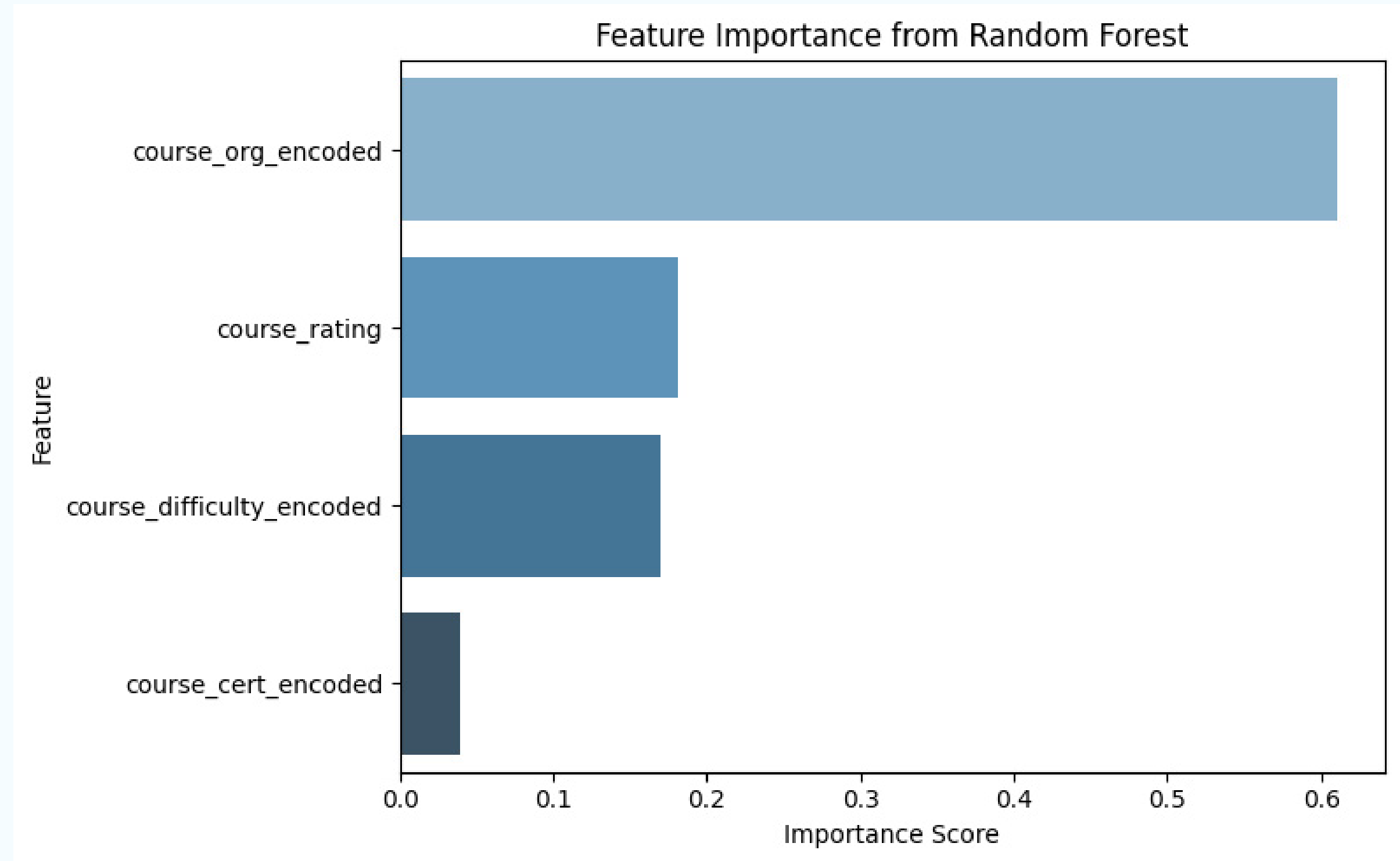


Output
Obtained



 Linear Regression Results:
Mean Squared Error (MSE): 12562082797.67
 R^2 Score: -0.034

Important Features



Insight:

"Although the course organization showed the strongest influence on enrollment, the overall model performed poorly. This suggests that student engagement is shaped by factors beyond visible course attributes — likely involving behavioral and contextual elements."

Recommendations

Mismatch Between Ratings and Engagement:

Courses with high ratings didn't always attract high enrollments.

Predictive models like Random Forest and Linear Regression had low R^2 scores, this reinforces the need to look beyond surface-level metrics and consider:

Course description quality

Instructor engagement

Marketing visibility

Social proof or peer influence



Root Cause Analysis, Key Conclusions

Mismatch Identified:

Highly-rated courses often had low student enrollments, indicating that quality doesn't always drive popularity.



Deeper Root Causes Explored:



- **Difficulty Level Bias:**

Many underperformers were Intermediate or Advanced.
Learners may avoid complex content, even if rated highly.

- **Organization Influence:**

Courses from prestigious institutions had high ratings but lower reach.

Suggests a gap between academic credibility and mass learner needs.

- **Certificate Requirements:**

Courses requiring paid certificates showed reduced enrollment.

Indicates cost-sensitivity or lower perceived value.

